1-Pager: LuatCheck

A Local, Evidence-Grounded Vietnamese Legal Assistant

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PROBLEM

Public legal chatbots may expose sensitive data and generate unsupported legal claims. LuatCheck is a privacy-sensitive Vietnamese legal assistant that retrieves relevant legal provisions, verifies whether claims are supported, and provides traceable citations. When evidence is insufficient or the question is ambiguous, it asks for clarification or refuses to answer.

The project combines local or organization-hosted deployment, evidence-grounded generation, clarification and refusal, controlled tool use, and security-aware permissions. It does not propose a new foundation model or retriever.

USERS

The target users are Vietnamese non-lawyers, especially employees, small-business owners, and SME administrative or HR staff who need understandable answers with inspectable legal evidence. Three representative situations:

1. An employee facing termination checks the notice period she is entitled to, and receives the answer together with the supporting provision.
2. SME HR staff ask a sensitive employment question without uploading the contract to a public chatbot.
3. A user pastes an untrusted document containing injected instructions, or asks how to evade a legal obligation, testing the security boundary and the refusal path.

WHY AN AGENT

Legal questions may require query reformulation, additional retrieval, clarification, evidence verification, answering, or refusal. A fixed workflow cannot determine the correct action for every question.

LuatCheck uses a bounded ReAct-style loop and a structured case ledger containing user facts, searches, retrieved article IDs, and the remaining tool budget. `search_law` and `get_article` are read-only; `save_case` is reversible; and `send_summary` requires explicit human confirmation. The agent terminates with an answer, clarification request, or refusal.

CANDIDATE AXES

The project will integrate:

- Planning and search: Hybrid BM25 and dense retrieval, RRF, reranking, and query reformulation.
- Reflection and verification: Claim-level citation checks, targeted repair, and refusal when evidence is insufficient.
- Guardrails and security: Untrusted-data isolation, permissioned tools, hard budgets, controlled egress, human confirmation, and audit logs.
- Memory: A short-term case ledger preserving facts, searches, article IDs, and budget. A long-term SQLite store keeps verified outcomes.
- Evaluation: Recall@5, Recall@20, MRR, unsupported-sentence rate, answer coverage, citation precision and recall, attack-success rate, and latency.